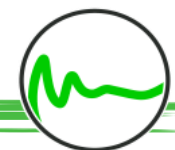
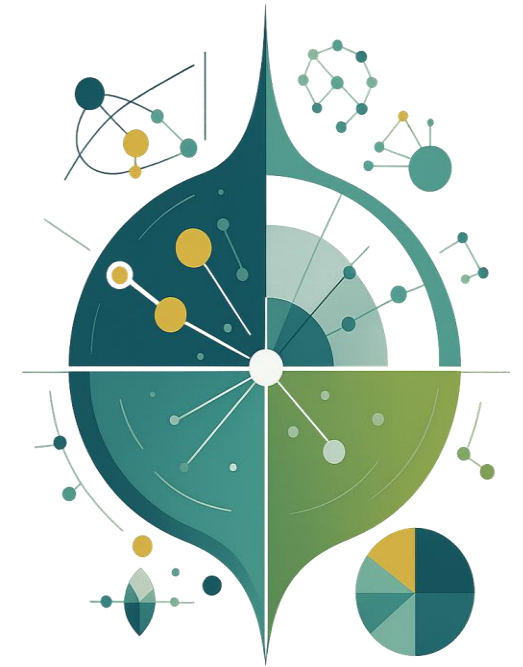


Welcome to the Science School on Quantitative Ecology!

Species Distribution Modelling

Dr. Anna Walentowitz

anna.walentowitz@uni-bayreuth.de



biogeografie
uni bayreuth

Bayceer
Bayreuth Center of Ecology
and Environmental Research



Aims

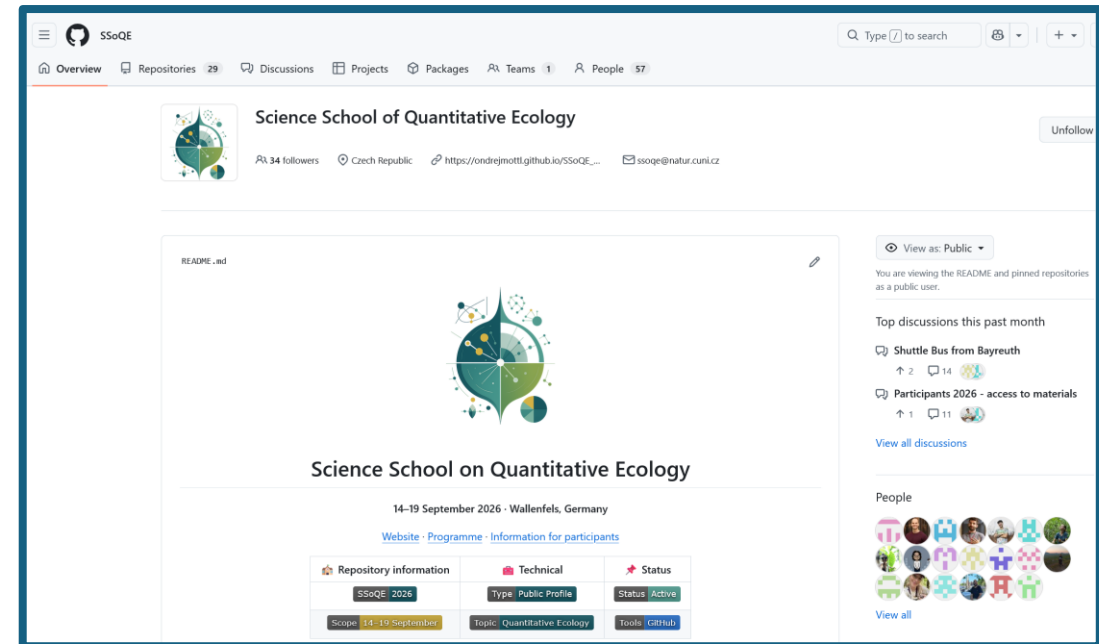
1. Understand the basic functioning and the application of SDMs.
2. Build a simple correlative SDM in R.
3. Generate a checklist for SMD building.
4. Collect resources for future SDM building.



Before we start:

Please check your setup:

Do you have access to the Science School materials on github?





Before we start:

Please check your setup:

Do you have R and R Studio installed?





Before we start:

Please check your setup:

Do you have the data and R script for the SDM session on your laptop?

The screenshot shows the GitHub interface for the repository 'SSoQE-Species_Distribution_Modelling'. The repository is private and has 1 branch and 6 tags. The main branch is selected. The file list shows the following files and their commit history:

File	Commit Message	Commit Time
.ai	Prepare species distribution repository rename (#1)	2 weeks ago
.cursor/rules	Prepare species distribution repository rename (#1)	2 weeks ago
.github	Prepare species distribution repository rename (#1)	2 weeks ago
data	Updated, change in readme	5 minutes ago
scripts	Updated, change in readme	5 minutes ago
.gitignore	Initial commit	2 years ago
AGENTS.md	Prepare species distribution repository rename (#1)	2 weeks ago
CLAUDE.md	Prepare species distribution repository rename (#1)	2 weeks ago
GEMINI.md	Prepare species distribution repository rename (#1)	2 weeks ago
README.md	Updated, change in readme	5 minutes ago
SSoQE-Species_Distribution_Modelling.Rproj	Prepare species distribution repository rename (#1)	2 weeks ago

The right sidebar shows the 'About' section with a description: 'In this session the following questions are tackled: What are species distribution models (SDMs)? And how can I build a simple correlative SDM in R?'. It also shows the 'Releases' section with the latest release 'SSoQE 2026 v3' and the 'Packages' section.



SDMs – what for?

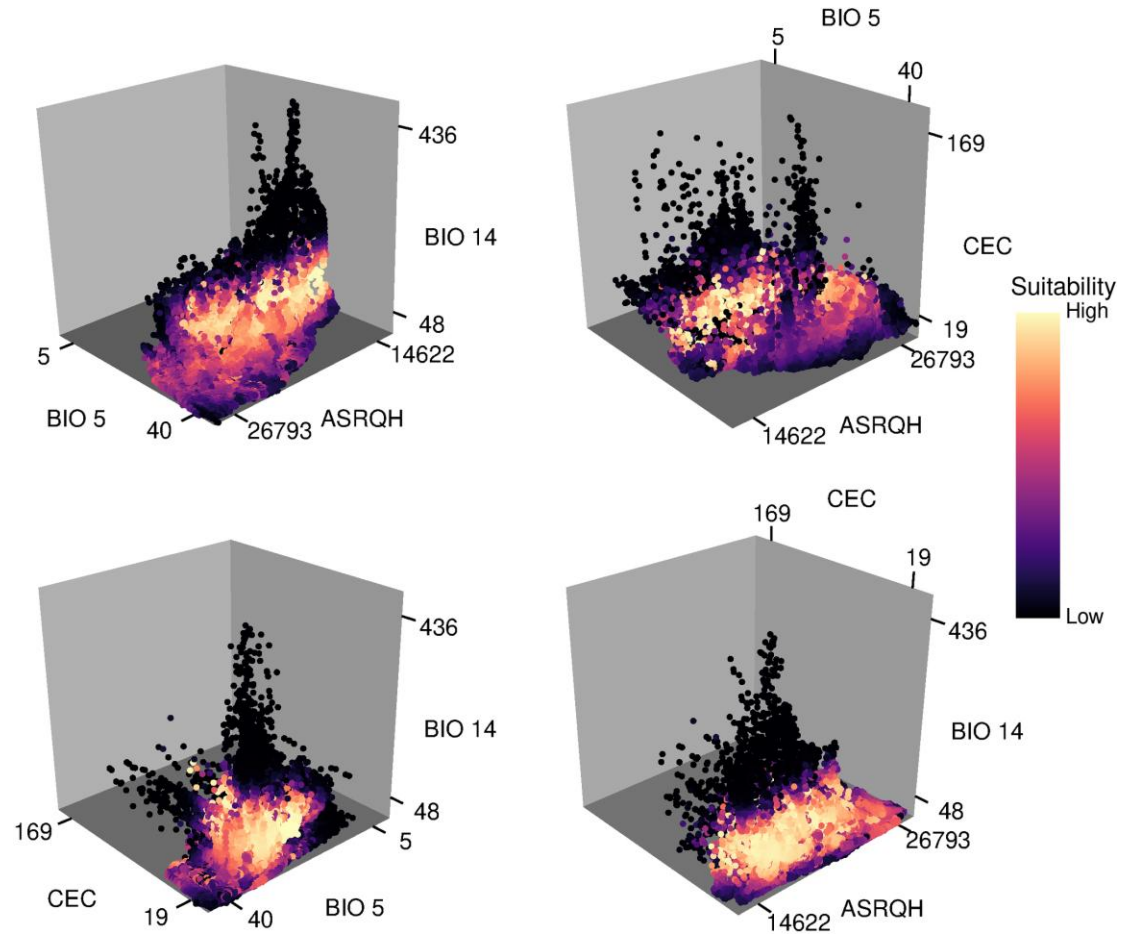


SDM – what for?

Understanding relationships

Here, species niches are examined.

Spatial predictions are not inherently necessary.

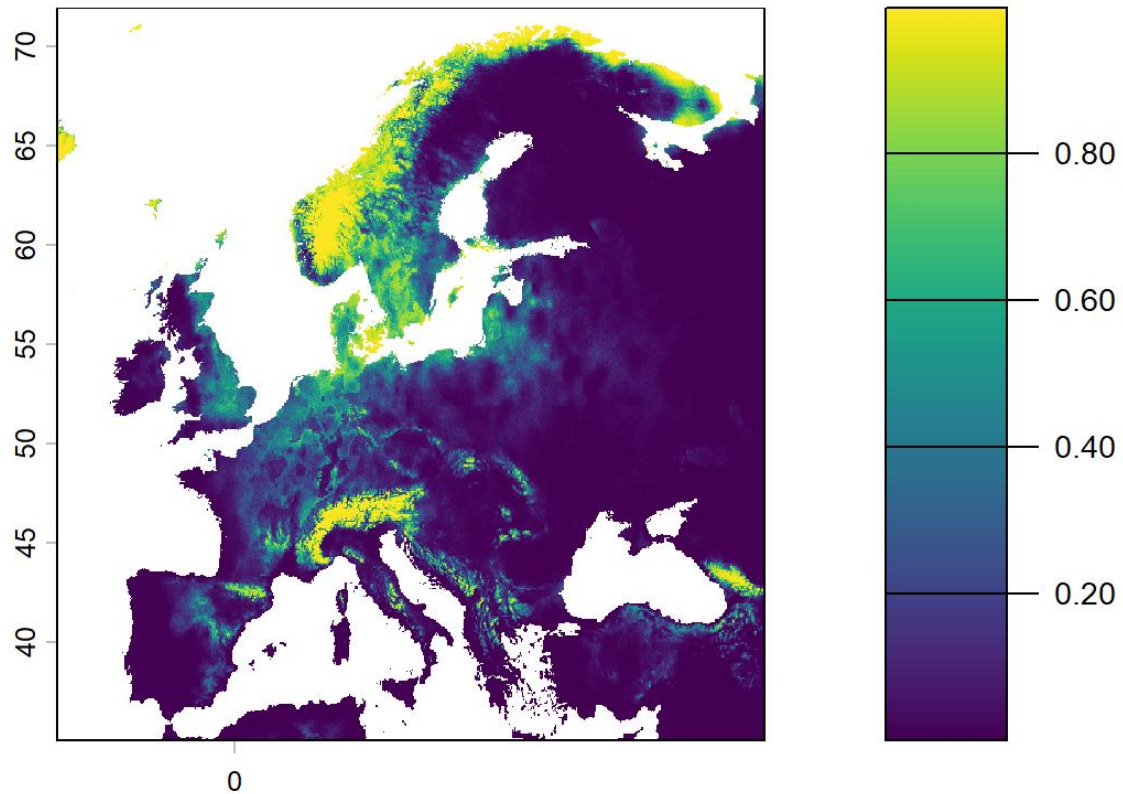


Cobos & Peterson, 2023



SDM – what for?

Predicting into the future



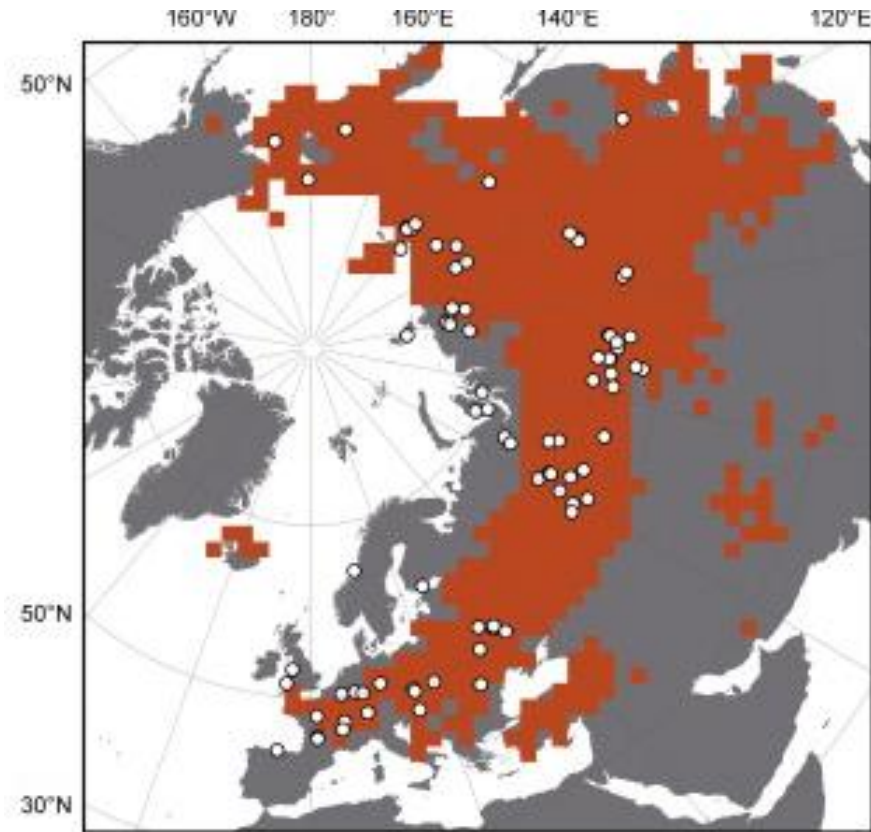
- Potential future distribution for *Arnica montana* under climate change

Output from SSoQE 2024



SDM – what for?

Predicting into the past



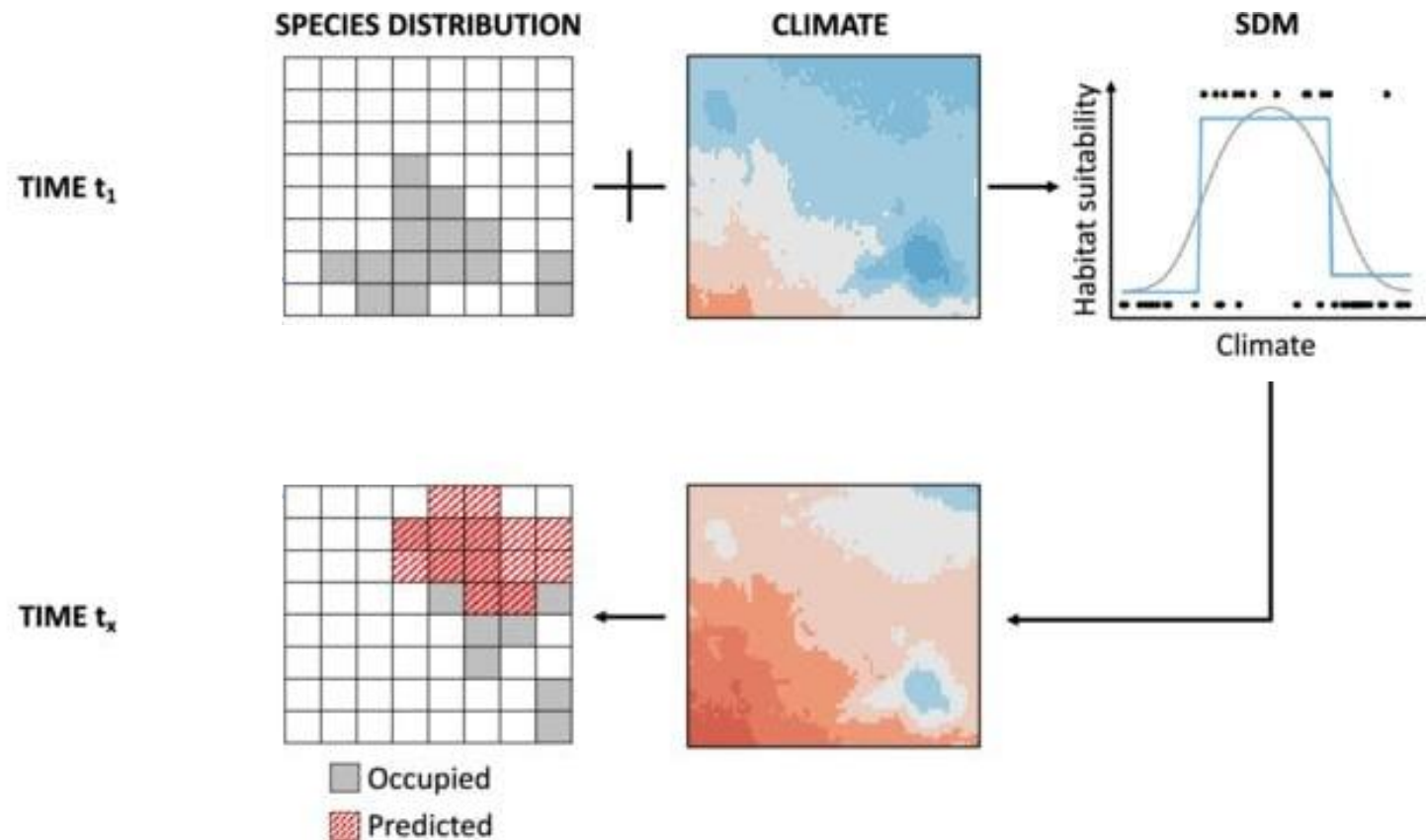
- SDM applied to palaeobiology
 - Here: SDM-estimated potential LGM distribution in Eurasia (red area) of woolly mammoth (*M. primigenius*) using fossils (dots) dated to the LGM (c. 21,000 ka BP)

Svenning et al., 2011



The variability of SDMs - examples

Correlative SDMs



- Conceptual overview of correlative species distribution models (SDMs) used for prediction under climate change.

adapted from Zurell et al., 2023



The variability of SDMs - examples

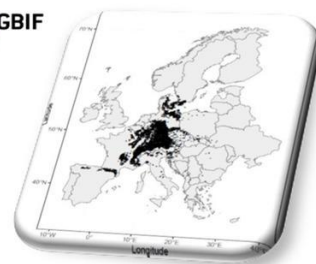
Ensemble SDMs

71 herbaceous plant species with affinity to *Fagus sylvatica* forests

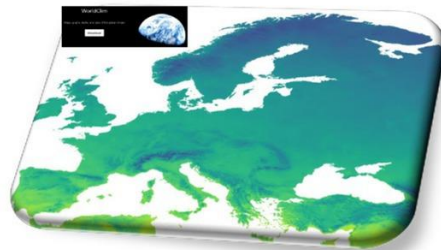


Species distribution modelling (ensemble approach)

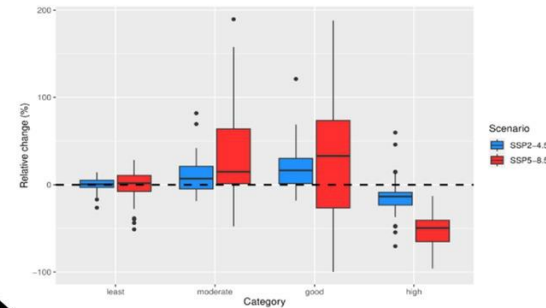
Occurrence records from GBIF database



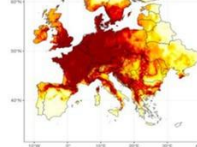
Bioclimatic variables from WorldClim database



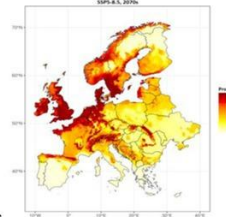
Climatic suitability is predicted to decrease in the future



strong northward shift



changes depend on biogeographic attributes of species



- Climatic suitability for European beech forest herbs under climate change

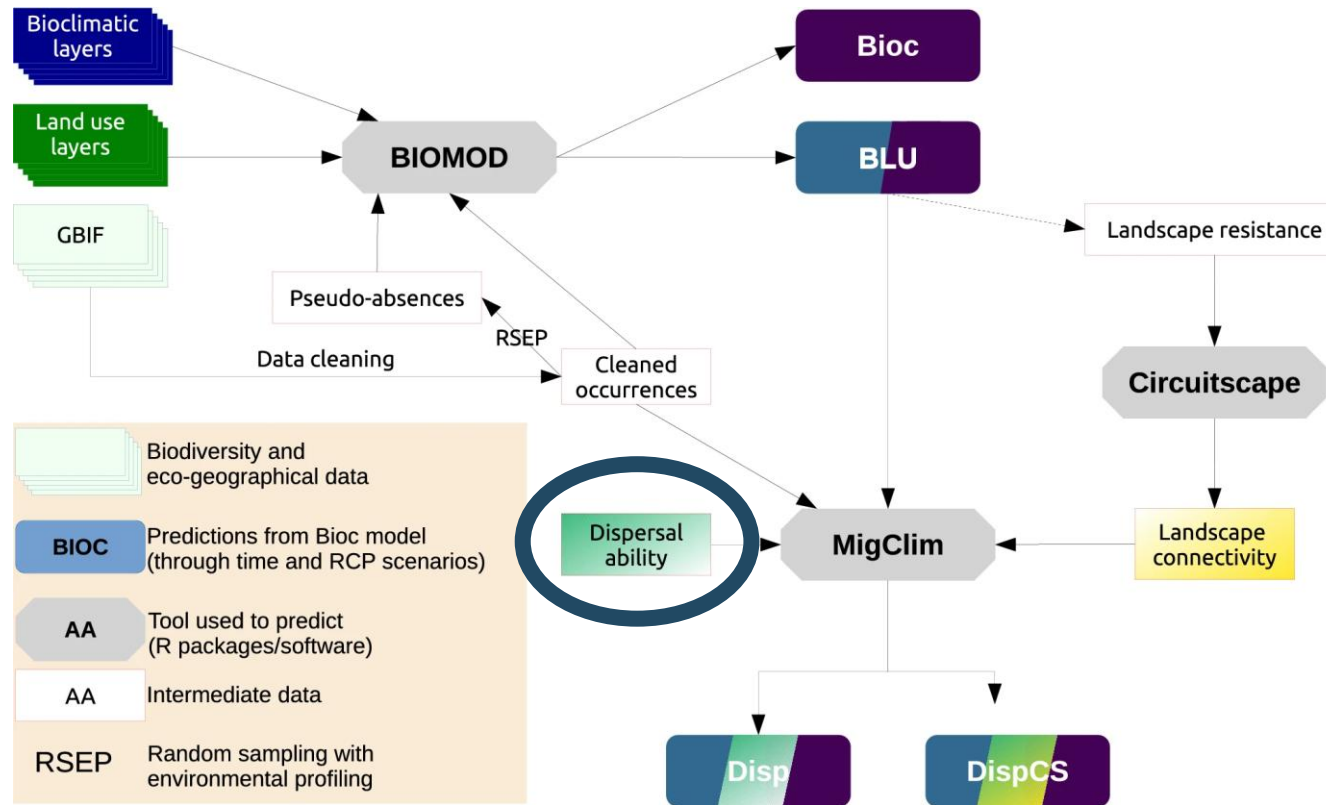
Kermavnar et al., 2023



The variability of SDMs - examples

Extended correlative SDMs

- Predictions with SDM of predatory arthropods in Europe

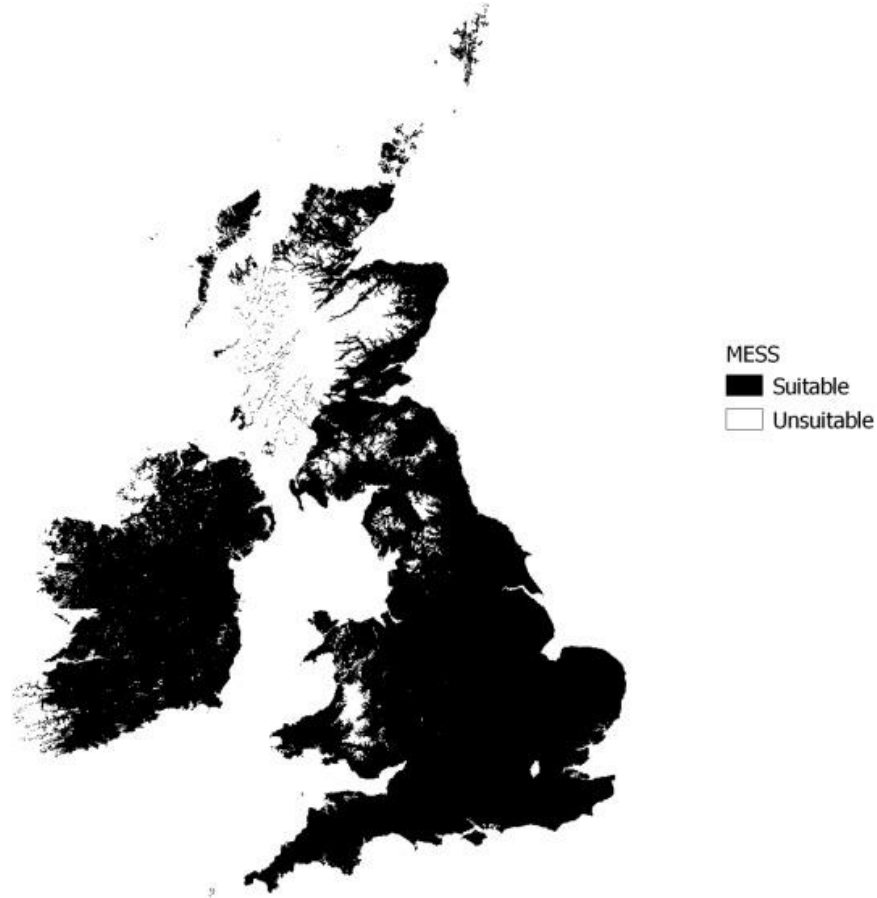


Monsimet et al., 2020



The variability of SDMs - examples

invasive (correlative) species distribution model (iSDM)



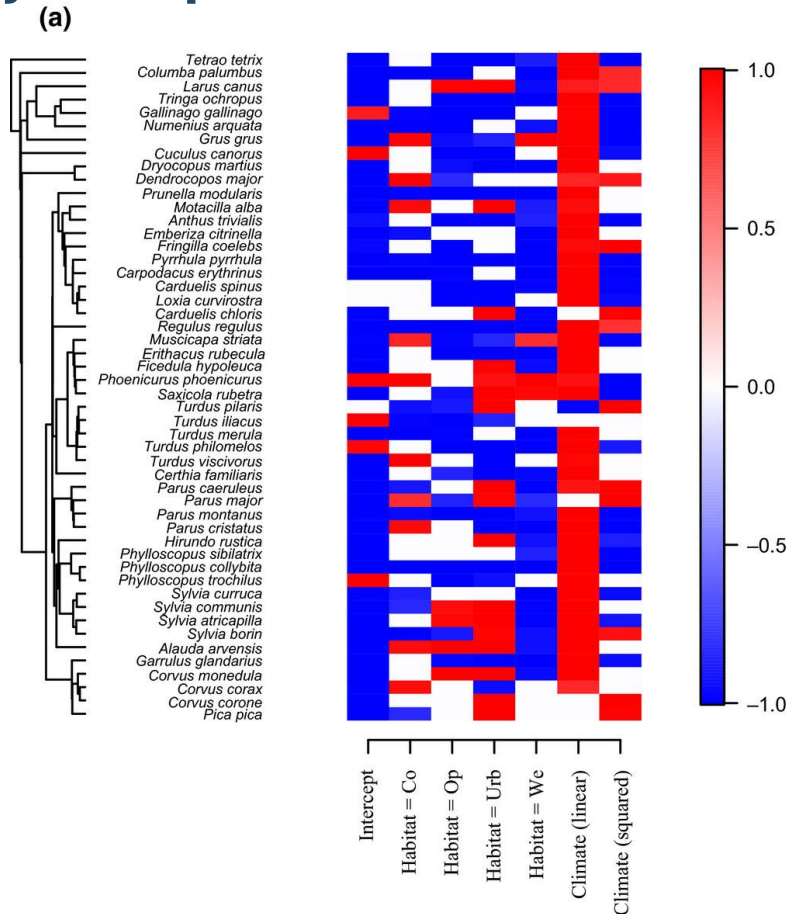
- Predicting the range of muntjac deer in Britain and Ireland
- Incorporation of non-equilibrium bias and survey effort in presence-only iSDM

Freeman et al., 2022



The variability of SDMs - examples

joint species distribution model (jSDM)



Includes species responses to environmental covariates...

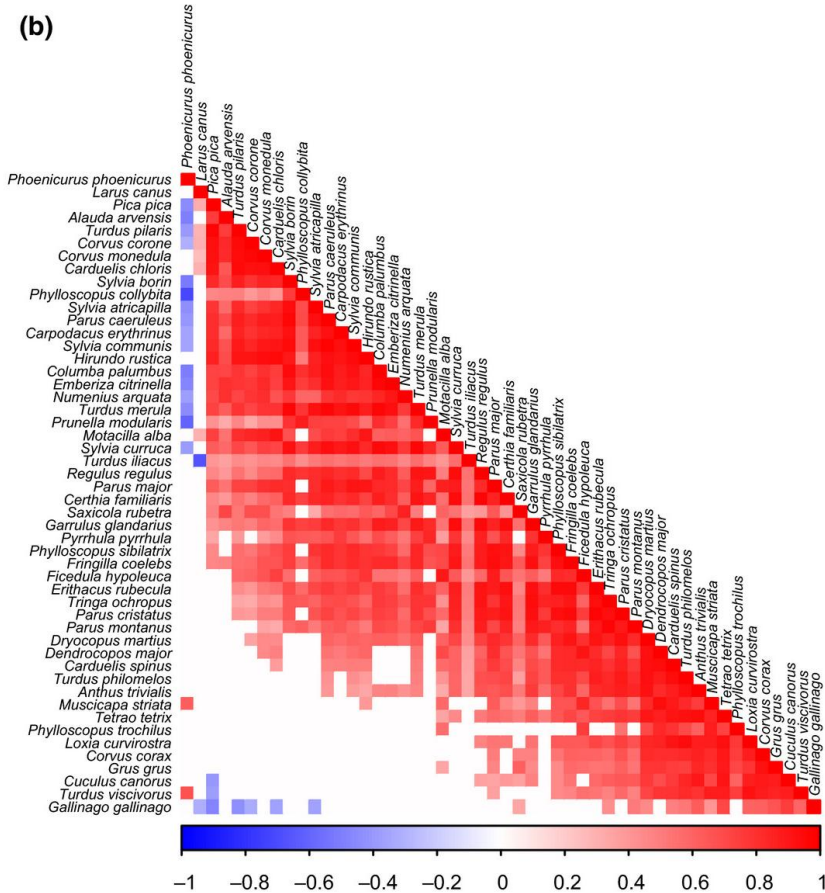
Tikhonov et al., 2019



The variability of SDMs - examples

jSDM

(b)



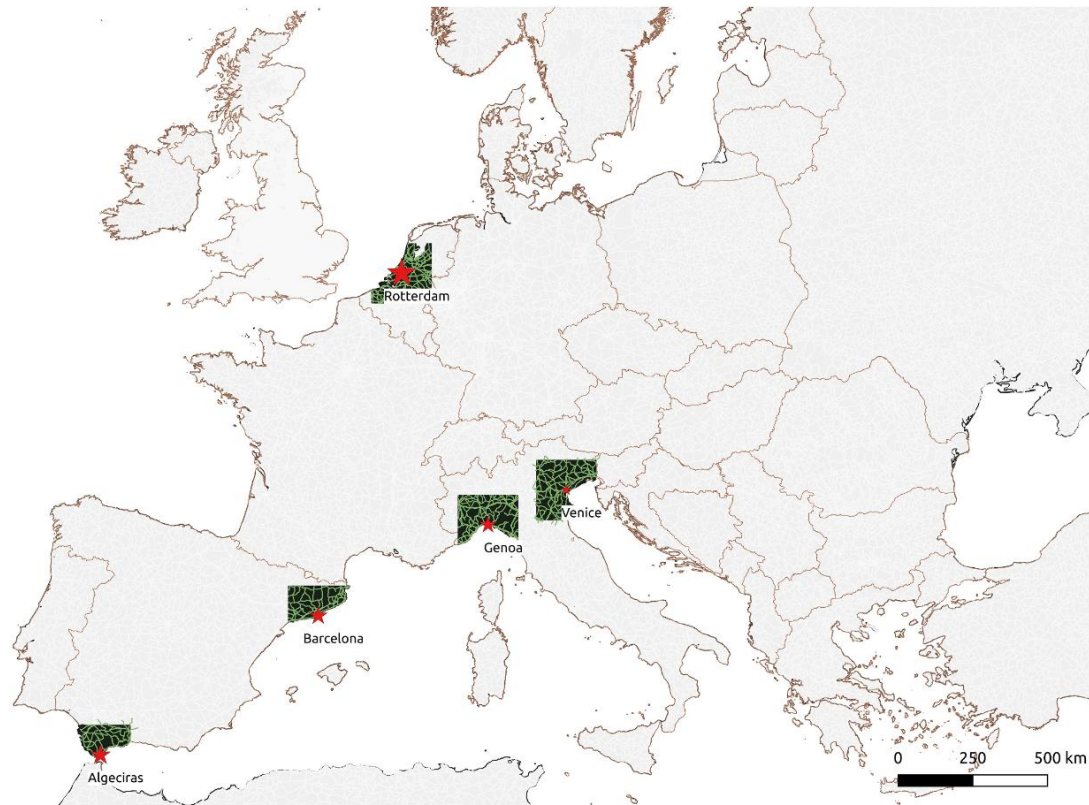
... and species' associations based on residuals.

Tikhonov et al., 2019



The variability of SDMs - examples

process-based SDMs



Aedes aegypti



Modeling of *Ae. aegypti* population dynamics in European ports.

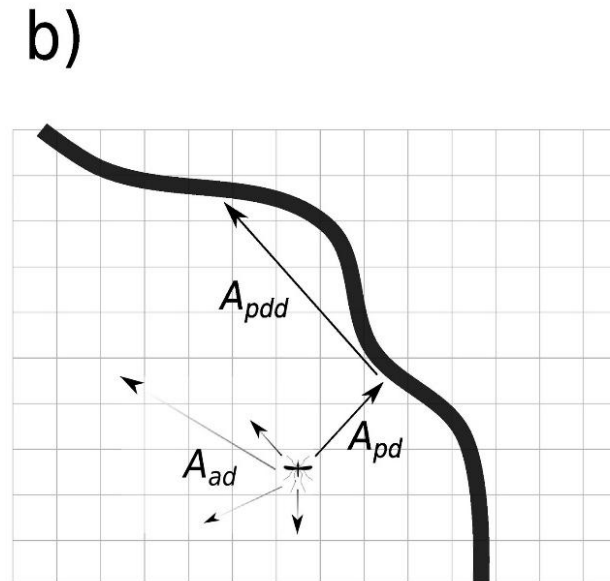
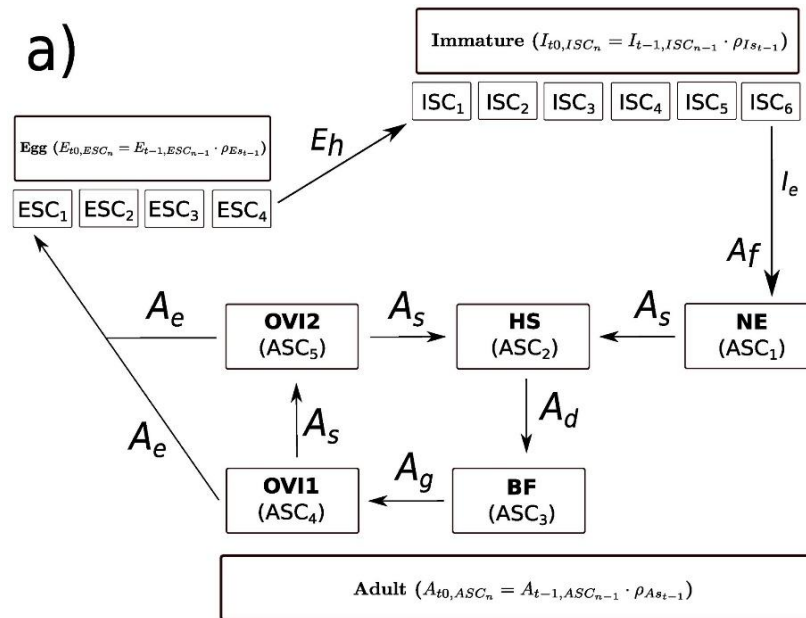
Ae. aegypti is a potential vector of yellow fever.

Da Re et al., 2021



The variability of SDMs - examples

process-based SDMs



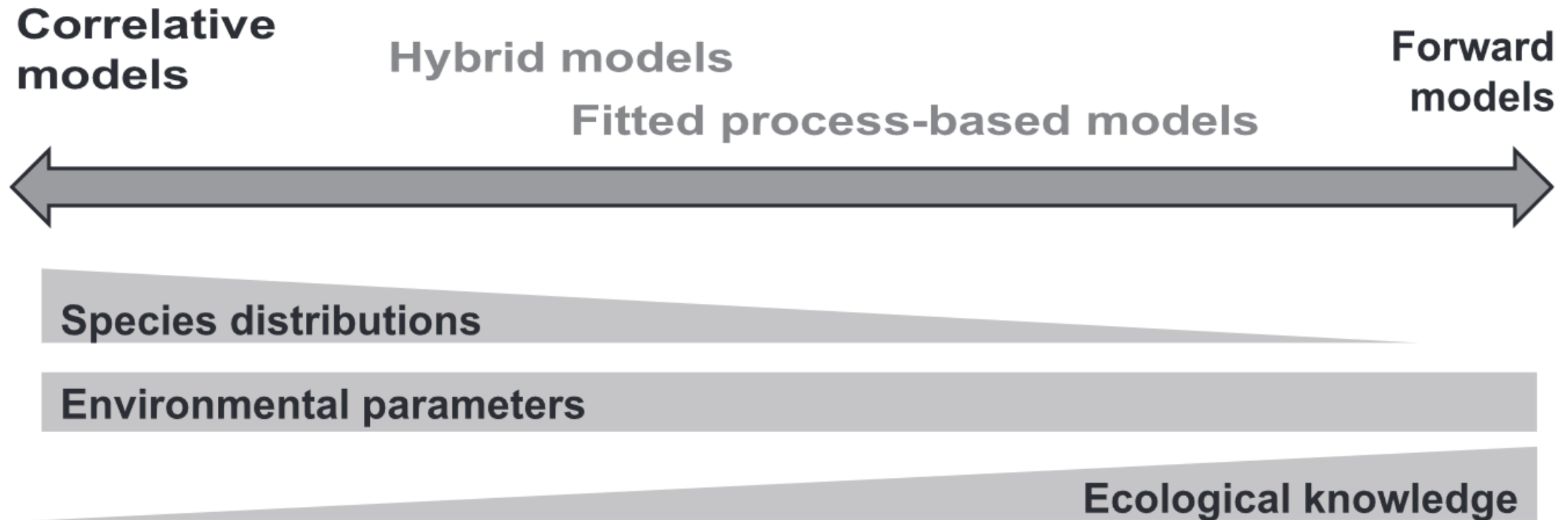
Modeling the life cycle...

... and dispersal.

Da Re et al., 2021



Correlative vs. process-based models

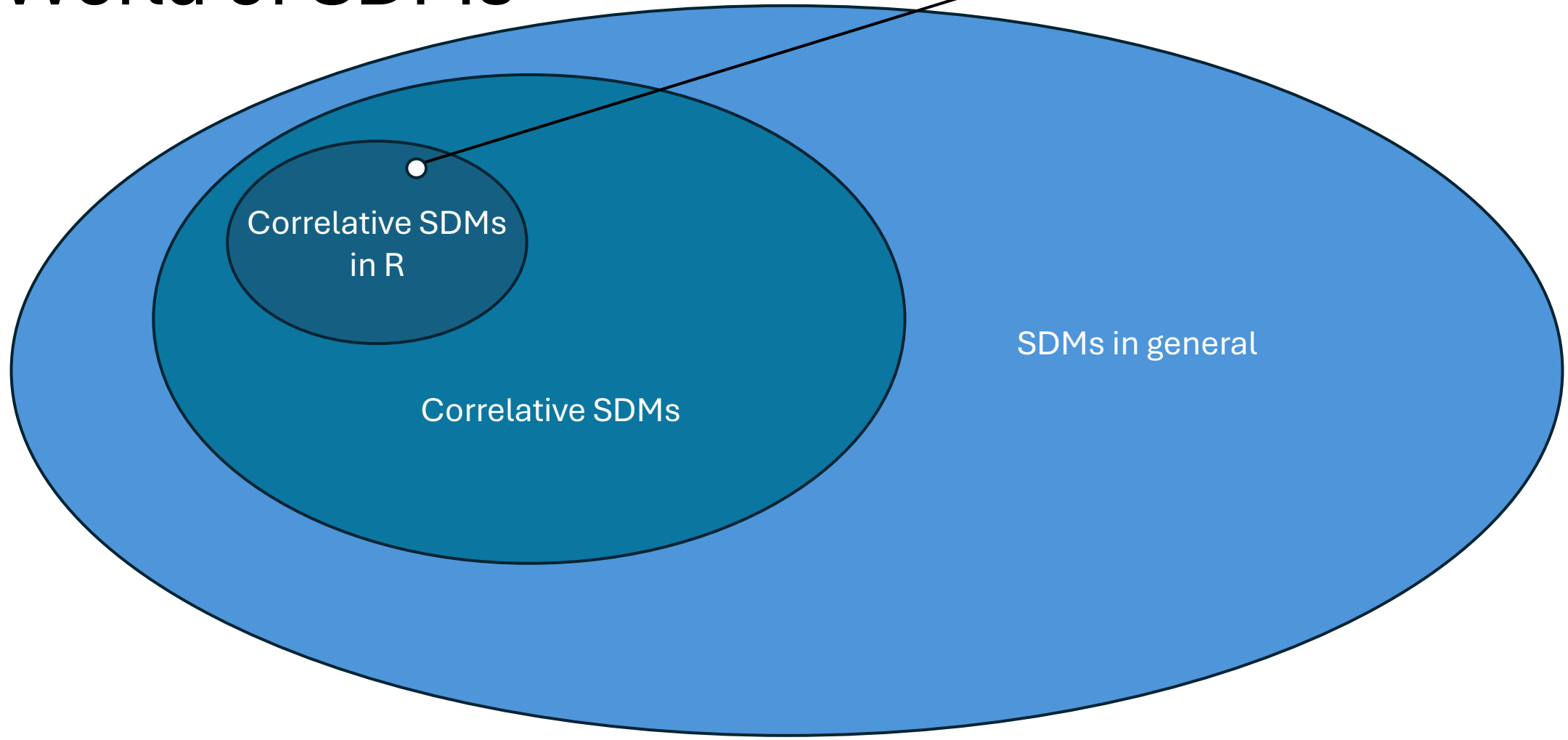


Dormann et al., 2011



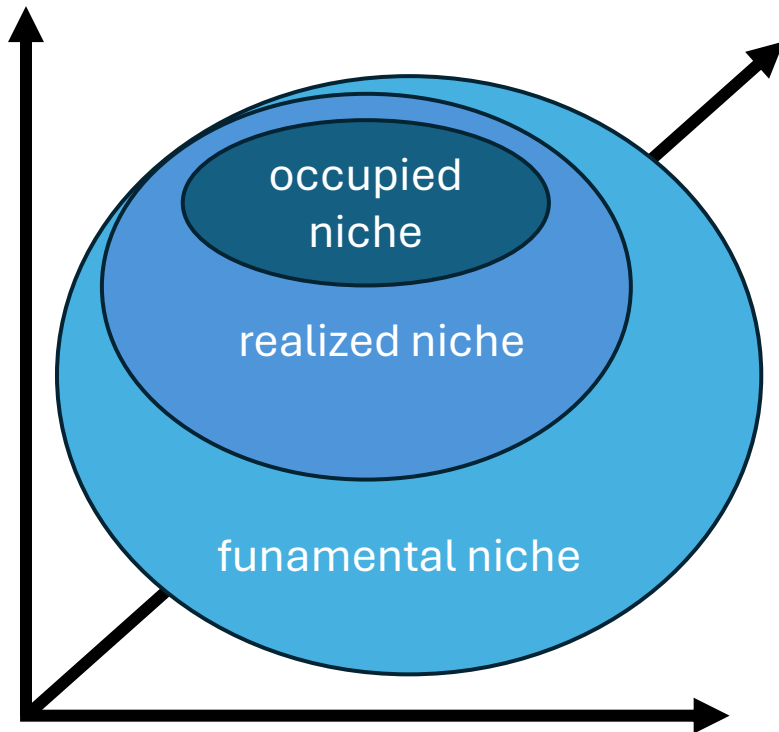
World of SDMs

What we will cover today





Underlying niche theory



- SDMs are rooted in niche theory
- **Fundamental niche:** abiotic conditions under which a species survives
- **Realised niche:** species survives despite presence of competitors
- **Occupied niche:** Presence of species considering dispersal limitations
- Ongoing debate which niche SDMs represent
- **Be aware what your model represents!**

Zurell, 2020

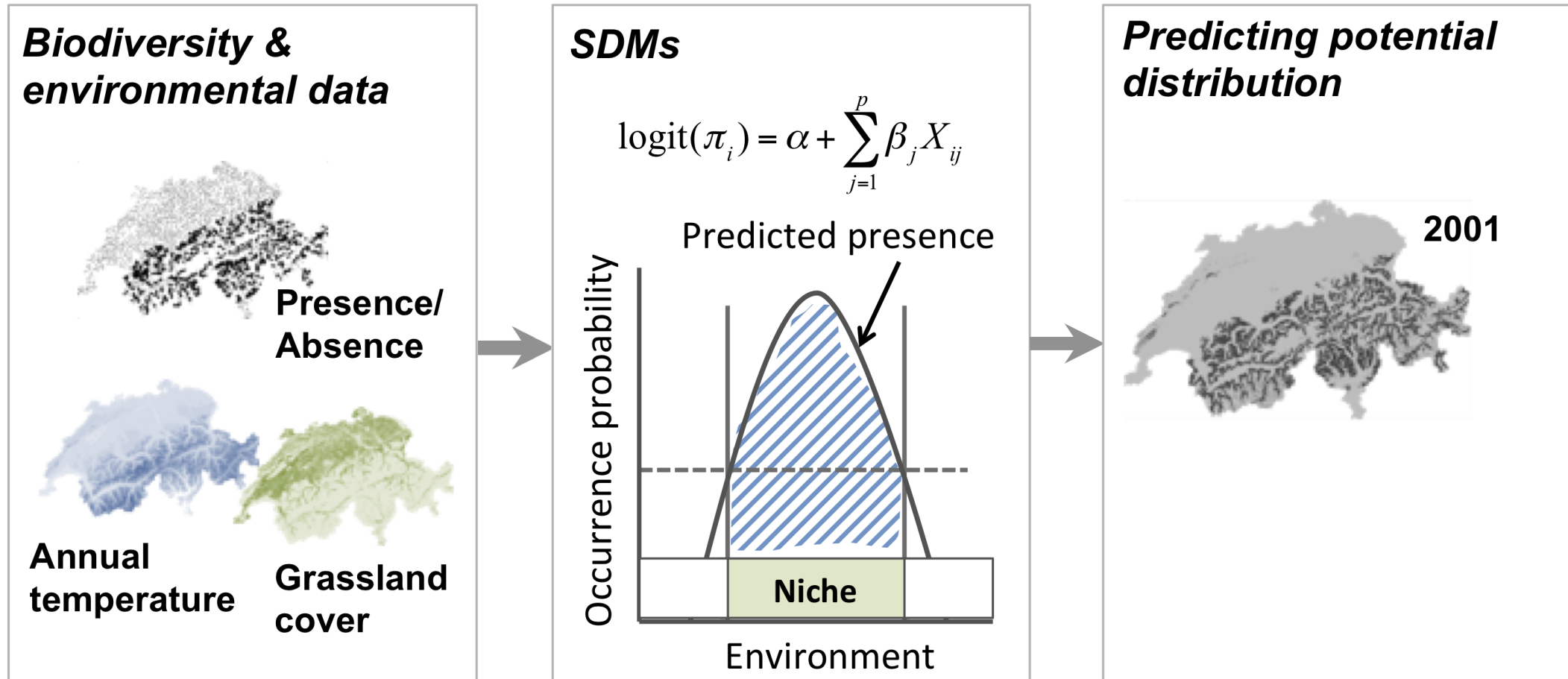
Questions?



Building a correlative SDM



Our goal: build simple correlative SDM



modified from Zurell, 2022



Checklist for building a (correlative) SDM



Your turn!

- **Goal:** Compile a checklist containing all aspects to build a scientifically sound and ecologically meaningful SDM.
- **Categories in the checklist:**
 - A. Occurrence and environmental data
 - B. Model building & quality check
 - C. Projections, visualization, reporting
- **Part 1:** Take 10 minutes to think about items to add to the checklist in your category
- **Part 2:** In a group with an A, B and C of your choice: compile a complete checklist (in 5 to 7 minutes)
- **Part 3:** In plenary, we compile the ultimate checklist for SDM building.



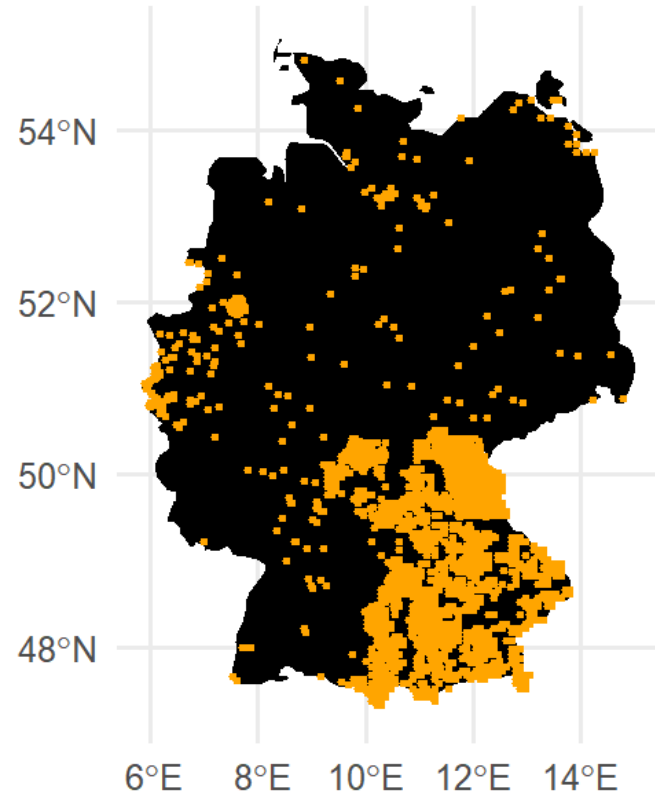
Species occurrence data

- **Data cleaning:** Get familiar with your data to be able to:
 - remove duplicate coordinates
 - be aware of missing values
 - be aware of uncertainties (e.g. coordinate uncertainty)
 - delete location errors (e.g., terrestrial species in the ocean).
- **Taxonomic verification:** Is my study species known under synonyms?
- **Dealing with spatial bias:** Spatial bias might be introduced by sampling design. Use thinning of occurrence points so highly sampled areas don't artificially bias the model.



Species occurrence data: spatial bias

Occurrences in Germany



Solutions:

- thinning
- filtering

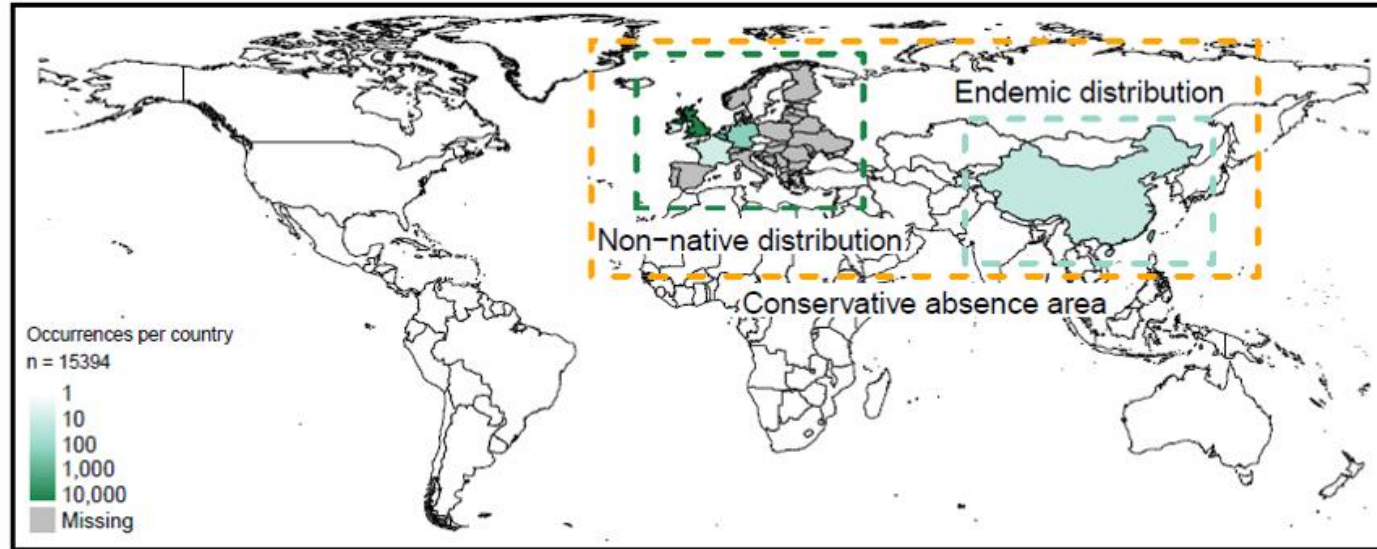


Species occurrence data

- **Data cleaning:** Get familiar with your data to be able to:
 - remove duplicate coordinates
 - be aware of missing values
 - be aware of uncertainties (e.g. coordinate uncertainty)
 - delete location errors (e.g., terrestrial species in the ocean).
- **Taxonomic verification:** Is my study species known under synonyms?
- **Dealing with spatial bias:**
 - Spatial bias might be introduced by sampling design. Use thinning of occurrence points so highly sampled areas don't artificially bias the model.
 - Select meaningful study area.



Species occurrence data: spatial bias



Solutions:

- Be aware of the study area you use. Should fit to your aim.

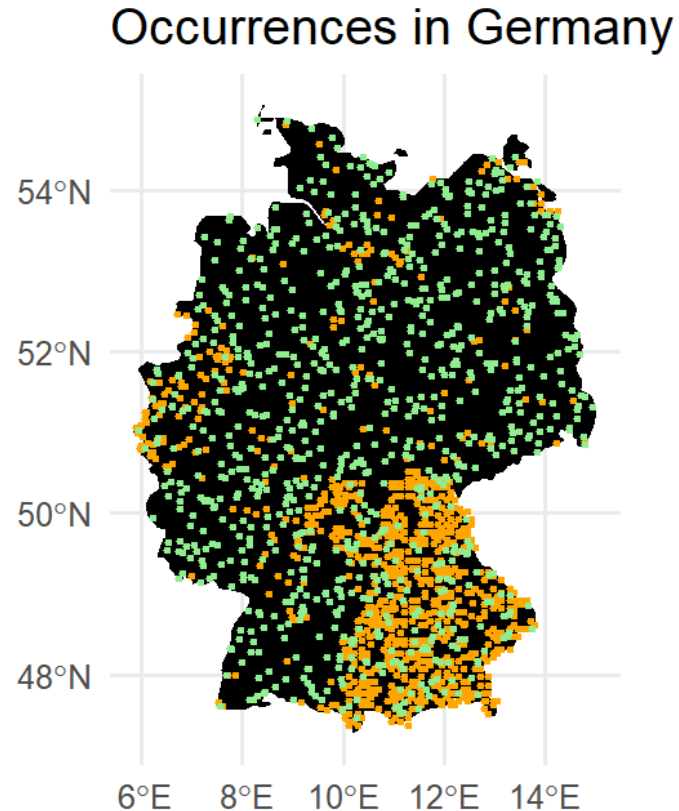


Species occurrence data

- **Consider time:** Make sure the dates of your occurrences matches the timeframe of your environmental data.
- **Creating pseudo-absences:** If using presence-only data (like Maxent), define a clear strategy for generating background points or pseudo-absences (e.g., constraining them to accessible geographic regions).



Species occurrence data: pseudo-absences



Solutions:

- wise selection of strategy to generate pseudo-absences
- awareness: do you need – pseudo-absences or background points



Environmental data

- **Ecologically meaningful parameters:** Select variables (e.g., temperature, precipitation, soil type) based on the known physiology and ecology of the species, not just data availability or easy-accessibility of data.
- Spatial resolution and area of interest:
 - Check if the resolution of the data makes sense for the aim of the study.
 - Area of interest should be chosen based on aim of the study and species' characteristics.
 - Resolution and extent of all environmental data should match.
 - Be aware of NAs in your data.
- Correlation: Test correlation of environmental parameters and remove highly correlated variables (typically $r > 0.7$) to prevent over-fitting.



Model building

- **Selection of algorithm:** Choose algorithm that fits your data study aim and species.
- **Data splitting:** Split data into training and testing sets (e.g., 70/30 split) or use k-fold cross-validation to build a model on part of the data but use the rest of the data to check your model. Spatial block validation is also a solution for clustered data where spatial block are used for cross-validation instead of randomly splitting the data, accounting for spatial autocorrelation.
- **Model finetuning:** Model settings can be finetuned and should fit the aim of the study and species even though it is tempting to use the default settings.

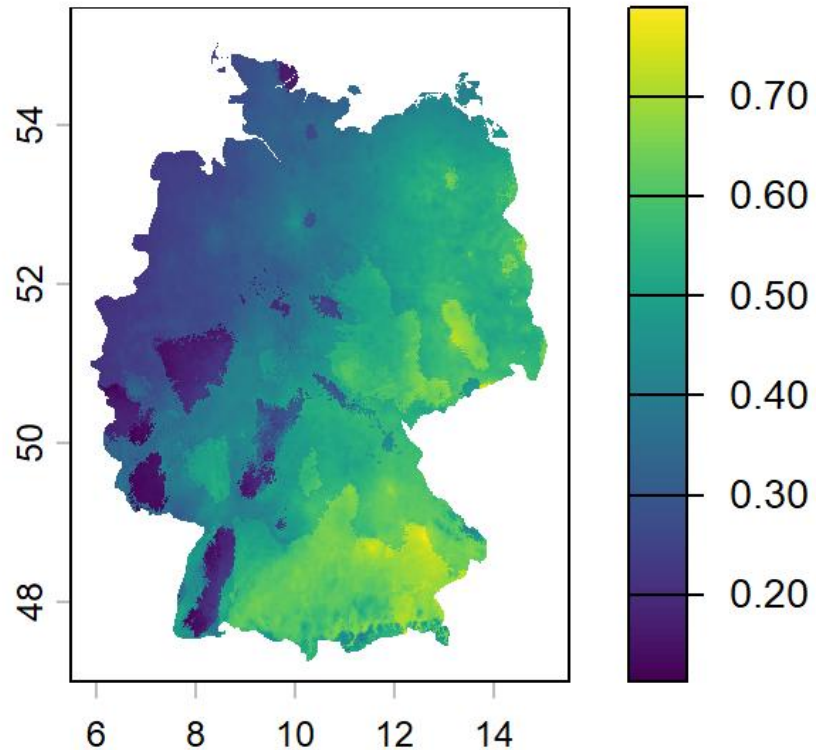


Model quality check

- **Metric to check model performance:** Evaluate the model using multiple metrics, such as AUC-ROC (checks sensitivity and specificity of the model), TSS (True Skill Statistic), Boyce Index (best for presence-only data).
- **Check if ecologically meaningful:** Inspect the response curves to ensure that the relationships between the species and the predictors make biological and ecologically sense.
- **Variable Importance:** Identify which environmental drivers contribute the most to the model's predictions and of that makes sense.
- **Iterate model building:** Model building is an iterative process. Build, check, revise. Check again revise again and so on.



Model quality check



Check based on your knowledge of the species:

- Might the patterns be driven by spatial bias in occurrence data?
- Might the environmental data be a poor fit for the model?



Projection and visualization

- **Generate suitability maps:** Generate continuous suitability maps to visualize spatial patterns of your SDM and, if necessary, apply a threshold to create binary presence/absence maps.
- **Clamping mask :** If projecting to the future (climate change) or other areas (invasive species), check for novel environmental conditions and apply clamping to prevent unrealistic extrapolations.
- **Embrace uncertainty:** showing not only the “nice” final maps but also a map depicting the variance or standard deviation study area add information on where uncertainty is higher and model outputs might not be very reliable.



Reporting

- **Reproducibility:**
 - Is my model building and analysis reproducible?
 - Did I properly cite data and literature or make own data accessible (see FAIRE & SHARE principles)?
 - If you have very long SI, that is usually a good sign.
- **Tables and Figures:** Did I report my model and model outcomes in an “honest” way without distorting patterns or exaggerating?



Reporting

- If done bad, this is what can happen to your paper:

Correction | [Open access](#) | Published: 01 May 2026

Correction: Reconstructing contact and a potential interbreeding geographical zone between Neanderthals and anatomically modern humans

[Saman H. Guran](#) , [Masoud Yousefi](#), [Anooshe Kafash](#) & [Elham Ghasidian](#)



Tutorials

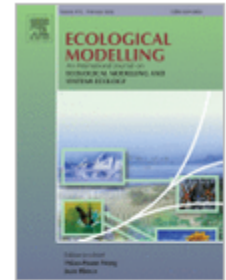
- **A very brief introduction to species distribution models in R** by Jeff Oliver:
<https://jcoliver.github.io/learn-r/011-species-distribution-models.html>
- **Introduction to species distribution modelling (SDM) in R** by Damaris Zurell:
<https://damariszurell.github.io/SDM-Intro/>
- **Downloading and cleaning GBIF data with R** by A.M. Barbosa: <https://www.r-bloggers.com/2021/03/downloading-and-cleaning-gbif-data-with-r/>
- **Accessing, handling, and referencing open biodiversity data using the Global Biodiversity Information Facility (GBIF)** by Eric Kusch:
<https://www.erikkusch.com/courses/gbif/>
- **Joint Species Distribution Modeling in R with Hmsc**:
<https://earthlab.colorado.edu/blog/joint-species-distribution-modeling-r-hmsc>

R packages



Ecological Modelling

Volume 476, February 2023, 110242



A curated list of R packages for ecological niche modelling

Neftalí Sillero ^a  , João Carlos Campos ^a, Salvador Arenas-Castro ^b, A.Márcia Barbosa ^a



Literature

Perspective | [Open access](#) | Published: 23 September 2019

A checklist for maximizing reproducibility of ecological niche models

[Xiao Feng](#) , [Daniel S. Park](#), [Cassandra Walker](#), [A. Townsend Peterson](#), [Cory Merow](#) & [Monica Papeş](#)

[Nature Ecology & Evolution](#) **3**, 1382–1395 (2019) | [Cite this article](#)

 [Save article](#)



Literature



Editor's Choice and Review and synthesis

 **Open Access**



A standard protocol for reporting species distribution models

[Damaris Zurell](#) , [Janet Franklin](#), [Christian König](#), [Phil J. Bouchet](#), [Carsten F. Dormann](#), [Jane Elith](#), [Guillermo Fandos](#), [Xiao Feng](#), [Gurutzeta Guillera-Arroita](#), [Antoine Guisan](#) ... **See all authors** 

First published: 01 June 2020 | <https://doi.org/10.1111/ecog.04960> |

 **VIEW METRICS**

[Go here for SFX](#) 

biogeografie
uni bayreuth

Questions?



Thank you!